

Module 6 — Geant4 Basics

Week 6 · Geometry, materials, physics lists

6.1 Geant4's mental model

Geant4 is not a simulation; it is a **toolkit** for building simulations. Every application must provide three mandatory things:

- **Detector construction.** A class derived from `G4VUserDetectorConstruction` that returns a world volume.
- **Physics list.** Which particles, processes, and models are active. Either a pre-built reference list (e.g. `FTFP_BERT`) or a custom class.
- **Primary generator action.** How to seed each event (particle gun, HepMC reader, ...).

Optional user actions plug into the **run**→**event**→**track**→**step** hierarchy. Most of the code you'll write is a subclass of `G4UserSteppingAction` or a `G4VSensitiveDetector`.

6.2 Volumes: Solid → Logical → Physical

Geant4 separates shape, material, and placement:

- A **solid** is pure geometry (`G4Box`, `G4Tubs`, `G4Sphere`, ...).
- A **logical volume** attaches a material and a sensitive detector to a solid.
- A **physical volume** places a logical volume in space (`G4PVPlacement`) or as a replica (`G4PVReplica`).

```
auto air = G4NistManager::Instance()->FindOrBuildMaterial("G4_AIR");
auto worldS = new G4Box("World", 5*m, 5*m, 5*m);
auto worldL = new G4LogicalVolume(worldS, air, "World");
auto worldP = new G4PVPlacement(nullptr, {}, worldL, "World",
nullptr, false, 0, true);
```

6.3 Materials

Use `G4NistManager` for standard materials (Pb, Fe, Cu, Scintillator, ...). Build compounds explicitly if needed:

```
auto man = G4NistManager::Instance();
auto lead = man->FindOrBuildMaterial("G4_Pb");
auto sci = man->FindOrBuildMaterial("G4_POLYSTYRENE"); // scint proxy
// Custom mix:
auto tungPowder = new G4Material("tungPowder", 10.0*g/cm3, 2);
tungPowder->AddMaterial(man->FindOrBuildMaterial("G4_W"), 0.9);
tungPowder->AddMaterial(epoxy, 0.1);
```

6.4 Physics lists

Pre-built lists cover >95% of use cases. For sPHENIX:

- FTFP_BERT — general-purpose, recommended default for HEP (Bertini cascade below ~5 GeV, FTF above).
- QGSP_BERT — LHC-era alternative; slightly different hadronic shapes.
- FTFP_BERT_HP — adds high-precision low-energy neutron transport (slower).

```
#include "FTFP_BERT.hh"
auto runManager = G4RunManagerFactory::CreateRunManager();
runManager->SetUserInitialization(new FTFP_BERT);
```

6.5 Particle gun — your first event

```
class MyGen : public G4VUserPrimaryGeneratorAction {
public:
  MyGen() { gun_ = new G4ParticleGun(1); }
  ~MyGen() override { delete gun_; }
  void GeneratePrimaries(G4Event* e) override {
    gun_->SetParticleDefinition(
      G4ParticleTable::GetParticleTable()->FindParticle("e-"));
    gun_->SetParticleEnergy(5.*GeV);
    gun_->SetParticlePosition({0,0,0});
    gun_->SetParticleMomentumDirection({0,0,1});
    gun_->GeneratePrimaryVertex(e);
  }
private:
  G4ParticleGun* gun_;
};
```

6.6 Running

A complete minimal program lives under \$G4INSTALL/share/Geant4-11.2/examples/basic/B1. Build with CMake:

```
mkdir B1-build && cd B1-build
cmake $G4INSTALL/share/Geant4-11.2/examples/basic/B1
make -j$(nproc)
./exampleB1 run1.mac
```

6.7 Exercises

- Fire 1000 electrons of 5 GeV into a 20 cm × 20 cm × 20 cm lead block; print total deposited energy.
- Replace lead with a layered Pb/scintillator sampling calorimeter (alternating 2 mm Pb, 4 mm scintillator × 20 layers).
- Vary the physics list between FTFP_BERT and QGSP_BERT; plot the energy-deposit distribution for 100 GeV pions.

6.8 Self-assessment

- [] I can explain Solid vs. Logical vs. Physical volumes.
- [] I know at least two reference physics lists and when to use each.
- [] I ran B1 and saw the default visualization.